

D466

Analyzing and Visualizing Data

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A hotel group with properties in nine cities wants to show its board how monthly occupancy changed over the last 24 months. The brand team recently refreshed the corporate color palette. Which chart is best?

- A. A line chart with months on the horizontal axis.
- B. A treemap of occupancy by city.
- C. A scatter plot of occupancy against month number.
- D. A pie chart with 24 slices.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

In Tableau, an analyst at a 200-location franchise drags Region to Columns and Sales to Rows. The data source refreshes nightly from a cloud warehouse. What does Tableau produce by default?

- A. A bar for each region showing COUNT of sales records.
- B. A line of sales over time.
- C. A bar for each region showing SUM(Sales).
- D. A single number for total sales.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A retailer keeps each quarter's orders in its own table, orders_q1 and orders_q2, with identical columns. The tables were set up by a contractor who has since left. The analyst needs one result with all orders from both quarters. What should she use?

- A. INNER JOIN orders_q1 to orders_q2 on order_id.
- B. UNION ALL of the two tables.
- C. Place the orders_q2 columns beside the orders_q1 columns.
- D. Query only orders_q2, since it is the newest.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Each order row needs to show the customer's region, which is stored in a separate customers table keyed by customer_id. What should the analyst use?

- A.** GROUP BY region on the orders table.
- B.** A UNION of orders and customers.
- C.** A WHERE filter on region.
- D.** A JOIN of orders to customers on customer_id.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A clinic network founded in 1987 reports no-shows by site: Site A had 20 no-shows out of 100 appointments; Site B had 30 out of 1,000. A manager with a nursing background asks for the network's overall no-show rate. What is it?

- A.** 23%
- B.** 11.5%
- C.** 50%
- D.** About 4.5%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Across 10 stores, AVG(Profit Ratio) is 18%, but SUM(Profit) / SUM(Sales) is 12%. Which is the true overall profit ratio?

- A.** 18%, because averages are always more accurate.
- B.** 12%, because the ratio of sums weights each store by its sales.
- C.** 15%, the midpoint of the two.
- D.** Both are equally valid overall ratios.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An analyst is building a dashboard for a hospital's chief nursing officer, who has five minutes each morning and needs to know whether any unit is short-staffed today. The analyst has 60 metrics available and a new Tableau Server license. What design is best?

- A. A focused view of today's staffing vs. target by unit, with shortages highlighted at the top.
- B. As many chart types as possible to show off the new server.
- C. All 60 metrics, so nothing is ever missed.
- D. A detailed table of last year's staffing by month.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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8.

An analyst at a streaming service with 3 million subscribers got 140 answers to an in-app pop-up that users chose to answer; 72% said they would pay more for ad-free viewing. The pop-up used the new brand font. How should she present this to executives?

- A. Report the 72% with its limits (140 self-selected responders) and recommend a representative test.
- B. Say that 72% of all subscribers will pay more.
- C. Never show a finding from a sample this small.
- D. Report 2.16 million subscribers willing to pay more.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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9.

A dashboard at a 12-branch credit union shows two bars: Branch A loans \$980,000 and Branch B loans \$1,020,000. The y-axis starts at \$960,000, so B's bar looks three times as tall as A's. Both branch managers joined in 2021. What is the problem?

- A. Nothing; the underlying numbers are accurate.
- B. The cut axis makes a difference of about 4% look like 200%; bars should start at zero.
- C. Branch B really has three times Branch A's loans.
- D. Nothing; axes should always zoom to the data range.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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10. [TRANSFER — reversed stem / different variable]

Branch A has \$980,000 in loans and Branch B has \$1,020,000. By what percent do B's loans exceed A's?

- A. 40%
- B. About 3.9%
- C. 200%
- D. About 4.1%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A subscription app's monthly churn fell from 8% to 6% after a redesign by a new product team. A finance memo called it "a 2% improvement." What is the percent change in churn?

- A. -75%
- B. -2%
- C. -25% (a 2-percentage-point drop)
- D. About -33%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A conversion rate was 5.0% and then rose by 20% in relative terms. What is the new rate?

- A. 5.2%
- B. 25.0%
- C. 6.0%
- D. 4.0%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A hotel group ~~with properties in nine cities~~ wants to show its board how monthly occupancy changed over the last 24 months. ~~The brand team recently refreshed the corporate color palette.~~ Which chart is best?

Target word: chart type **Decides:** A change over time.

A	Correct	Lines show trends over continuous time.
B	Camouflage	Answers the nine-cities detail, not the 24-month trend.
C	Adjacent category	Scatter plots are for relationships between two measures.
D	Confusion set	Pies show parts of a whole, not change over time.

2. Answer C Concept 2

In Tableau, an analyst at a ~~200-location~~ franchise drags Region to Columns and Sales to Rows. ~~The data source refreshes nightly from a cloud warehouse.~~ What does Tableau produce by default?

Target word: default view **Decides:** Region is a dimension; Sales is a measure summed by default.

A	Confusion set	The default aggregation for a measure is SUM, not COUNT.
B	Adjacent category	A trend line needs a date field.
C	Correct	The dimension splits the view; the measure is summed.
D	Dropped element	Ignores the Region dimension.

3. Answer B Concept 3

A retailer keeps each quarter's orders in its own table, orders_q1 and orders_q2, with identical columns. ~~The tables were set up by a contractor who has since left.~~ The analyst needs one result with all orders from both quarters. What should she use?

Target word: combine method **Decides:** Same columns, different rows: stack them.

A	Confusion set	Different orders will not match; the result would be nearly empty.
B	Correct	Stacks all rows from both quarters.
C	Inverted operation	Widens the table instead of stacking rows.
D	Dropped element	Leaves out the first quarter.

4. Answer D Concept 3 · transfer

Each order row needs to show the customer's region, which is stored in a separate customers table keyed by customer_id. What should the analyst use?

Target word: reverse: when to join **Decides:** Adding an attribute from another table by key is a join.

A	Adjacent category	Orders do not contain region to group by.
B	Confusion set	UNION stacks rows and needs matching columns.
C	Dropped element	Cannot filter on a column that is not in the table yet.
D	Correct	A join adds the region column to each matching order.

5. Answer D Concept 4

A clinic network founded in 1987 reports no-shows by site: Site A had 20 no-shows out of 100 appointments; Site B had 30 out of 1,000. A manager with a nursing background asks for the network's overall no-show rate. What is it?

Target word: overall rate **Decides:** Total no-shows / total appointments.

A	Inverted operation	Adds the two rates.
B	Dropped element	Averages 20% and 3%, dropping the size difference.
C	Wrong stem number	The count of no-shows, read as a percent.
D	Correct	$50 / 1,100 \approx 4.5\%$.

6. Answer B Concept 4 • transfer

Across 10 stores, AVG(Profit Ratio) is 18%, but SUM(Profit) / SUM(Sales) is 12%. Which is the true overall profit ratio?

Target word: which aggregate is correct **Decides:** Overall ratio weights by sales.

A	Absolute language	An average of ratios lets small stores count as much as large ones.
B	Correct	Large stores count in proportion to their sales.
C	Inverted operation	Averaging two methods does not fix either.
D	Confusion set	Only the ratio of sums is the overall ratio.

7. Answer A Concept 5

An analyst is building a dashboard for a hospital's chief nursing officer, who has five minutes each morning and needs to know whether any unit is short-staffed today. The analyst has 60 metrics available and a new Tableau Server license. What design is best?

Target word: dashboard design **Decides:** The user's one decision: which units are short today.

A	Correct	Puts the decision-critical metric first for a five-minute read.
B	Camouflage	The license does not change what the user needs.
C	Absolute language	Clutter hides the one answer she needs.
D	Adjacent category	Historical, not today's status.

8. Answer A Concept 6

An analyst at a streaming service with 3 million subscribers got 140 answers to an in-app pop-up that users chose to answer; 72% said they would pay more for ad-free viewing. The pop-up used the new brand font. How should she present this to executives?

Target word: presenting with limitations **Decides:** Small, self-selected sample.

A	Correct	States the finding honestly and shows what would confirm it.
B	Dropped element	Drops the sample limitation and overgeneralizes.
C	Absolute language	It can be shown as directional evidence with caveats.
D	Wrong stem number	Multiplies a biased sample rate by the full subscriber base.

9. Answer B Concept 7

A dashboard at a 12-branch credit union shows two bars: Branch A loans \$980,000 and Branch B loans \$1,020,000. The y-axis starts at \$960,000, so B's bar looks three times as tall as A's. ~~Both branch managers joined in 2021.~~ What is the problem?

Target word: visual flaw **Decides:** The axis does not start at zero.

A	Absolute language	Accurate numbers can still be shown misleadingly.
B	Correct	Bar length must be proportional to value.
C	Dropped element	Drops the axis baseline and reads heights as values.
D	Adjacent category	Zooming can suit line charts, but not bars.

10. Answer D Concept 7 · transfer

Branch A has \$980,000 in loans and Branch B has \$1,020,000. By what percent do B's loans exceed A's?

Target word: actual percent difference **Decides:** $\$40,000 / \$980,000$.

A	Wrong stem number	Misplaces the decimal on \$40,000.
B	Inverted operation	Divides by B's value instead of A's base.
C	Confusion set	The visual height gap on a truncated axis, not the data.
D	Correct	$40,000 / 980,000 \approx 4.1\%$.

11. Answer C Concept 8

A subscription app's monthly churn fell from 8% to 6% ~~after a redesign by a new product team.~~ A finance memo called it "a 2% improvement." What is the percent change in churn?

Target word: percent change **Decides:** $(6 - 8) / 8$.

A	Wrong stem number	$6 / 8$ is the ratio of new to old, not the change.
B	Confusion set	Two percentage points, not two percent.
C	Correct	$2 / 8 = 25\%$ relative decline.
D	Inverted operation	Divides by the new value, 6.

12. Answer C Concept 8 · transfer

A conversion rate was 5.0% and then rose by 20% in relative terms. What is the new rate?

Target word: solve for the new rate **Decides:** $5.0\% \times 1.20$.

A	Adjacent category	Adds 0.2 points, mixing up units.
B	Confusion set	Adds 20 percentage points.
C	Correct	$5.0 \times 1.2 = 6.0$.
D	Inverted operation	Applies a 20% decrease.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Chart choice	Line charts show change over time; bar charts compare categories; scatter plots show relationships between two measures.	Using pie or scatter charts for a time trend.	“Over the last N months.”
2	Dimensions and measures in Tableau	A dimension splits the view into categories; a measure is aggregated, SUM by default.	Expecting row-level counts or a trend without a date.	A category on one shelf and a number on the other.
3	UNION vs. JOIN	UNION stacks rows from tables with the same columns; JOIN adds columns by matching keys.	Joining tables that should be stacked.	Same columns, different time periods vs. extra attributes in another table.
4	Ratio of sums vs. average of ratios	An overall rate is total numerator / total denominator, not the average of group rates.	Averaging percentages from groups of very different sizes.	Rates for groups with very different bases.
5	Dashboard design for the audience	Show the few metrics the audience needs for its decision, most important first.	Adding every available metric.	A named user with a specific, time-limited need.
6	Stating data limitations	Present findings with their limits (sample size, self-selection, missing data) and do not generalize past them.	Projecting a biased sample onto the whole population.	“Users who chose to answer.”
7	Truncated axes mislead	Bar lengths must start at zero; a cut axis exaggerates differences.	Reading bar height ratios as data ratios.	An axis starting near the data values.
8	Percent change vs. percentage points	Percent change = (new – old) / old; the raw difference between two percentages is in percentage points.	Calling a 2-point drop a 2% change.	Two rates and the word “percent.”

Course scope (public description): Analyzing and Visualizing Data examines the principles and techniques used to effectively analyze data to answer questions and share data through the art of visualization. This course explores how to use formulas and functions to perform calculations, how to design and create visualizations and dashboards in Tableau, and how to build an effective data presentation that considers limitations associated with the data and best practices for audience considerations. This course provides students with an opportunity to demonstrate an understanding of what is involved in the conversion and formatting of data and apply the use of functions and syntax to create SQL queries for combining data from multiple database tables. The course D467: Exploring Data, which also includes Google certificate materials, is a prerequisite.